Week 1

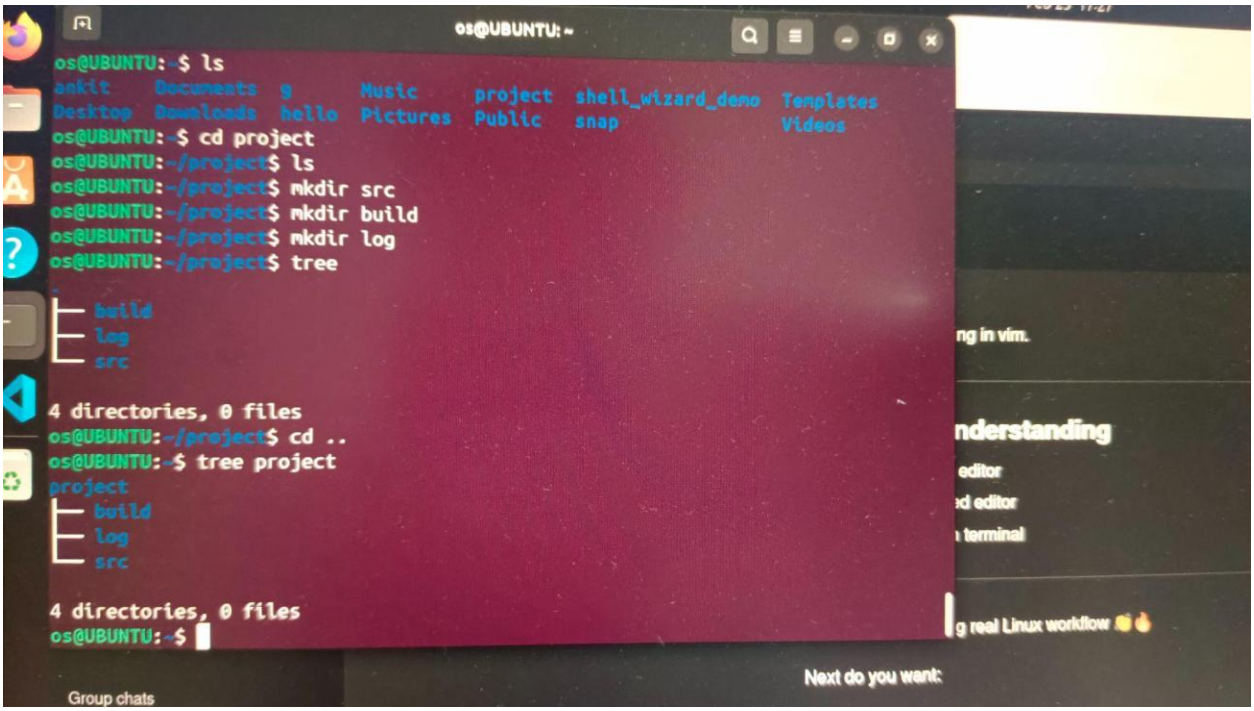
Assignment 1

Topic

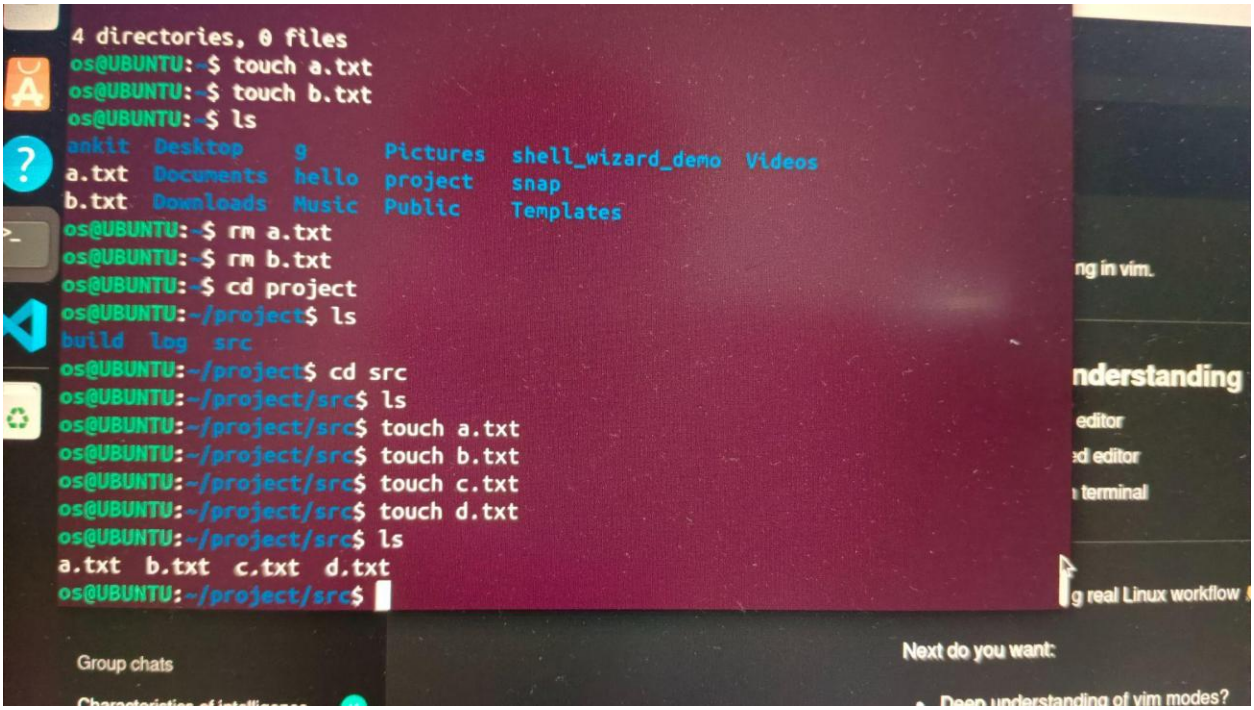
1. Create a directory structure like

```
|--project/
     |----src/
     |--build/
     |----log/
```

Ans

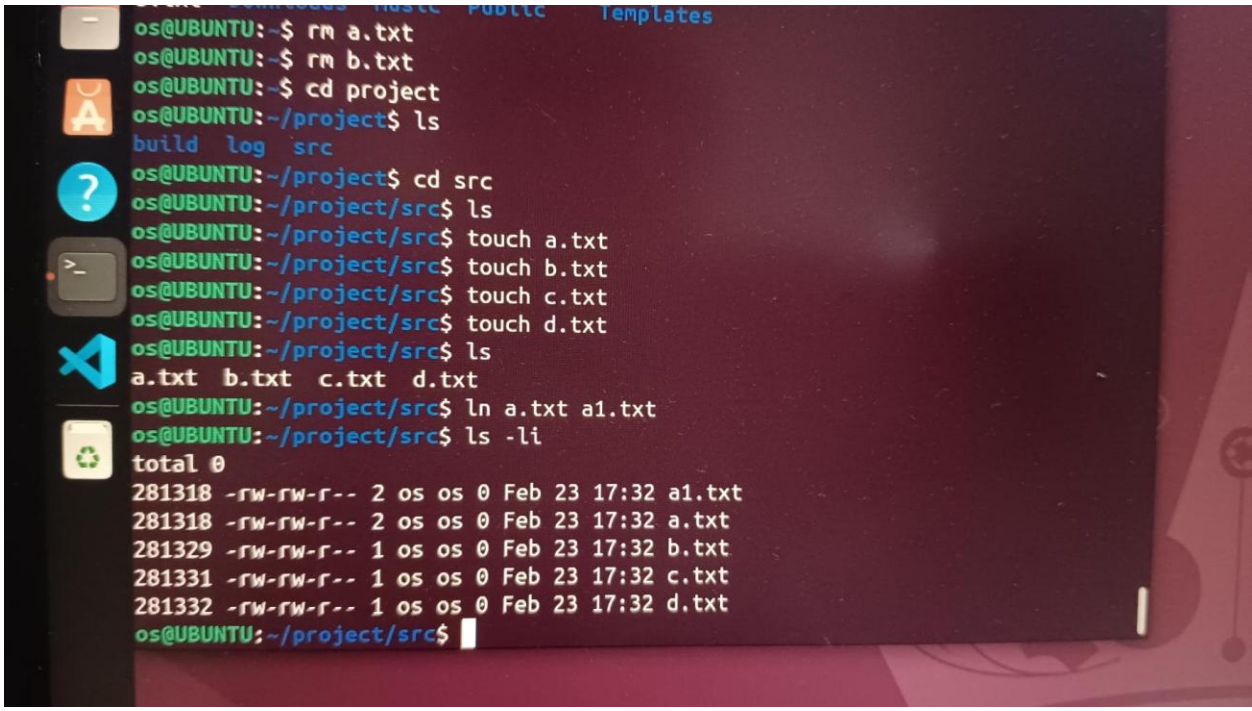


2. Create files inside `src/`

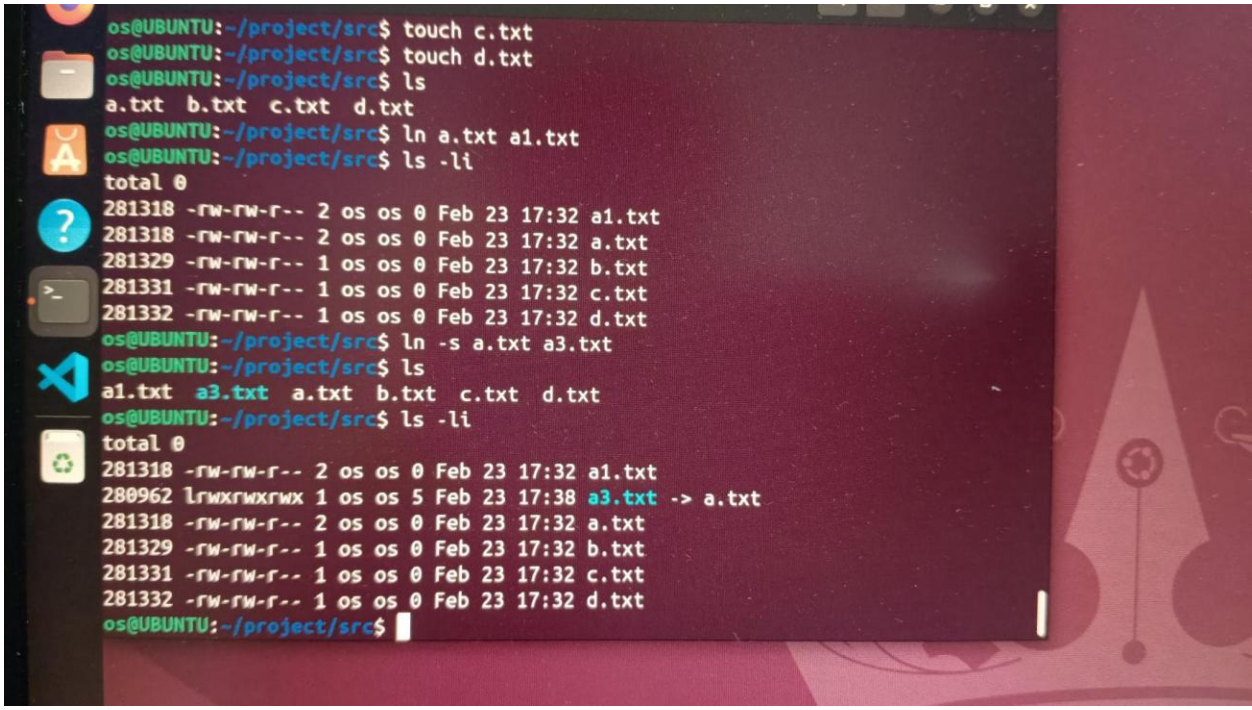


3. Create:

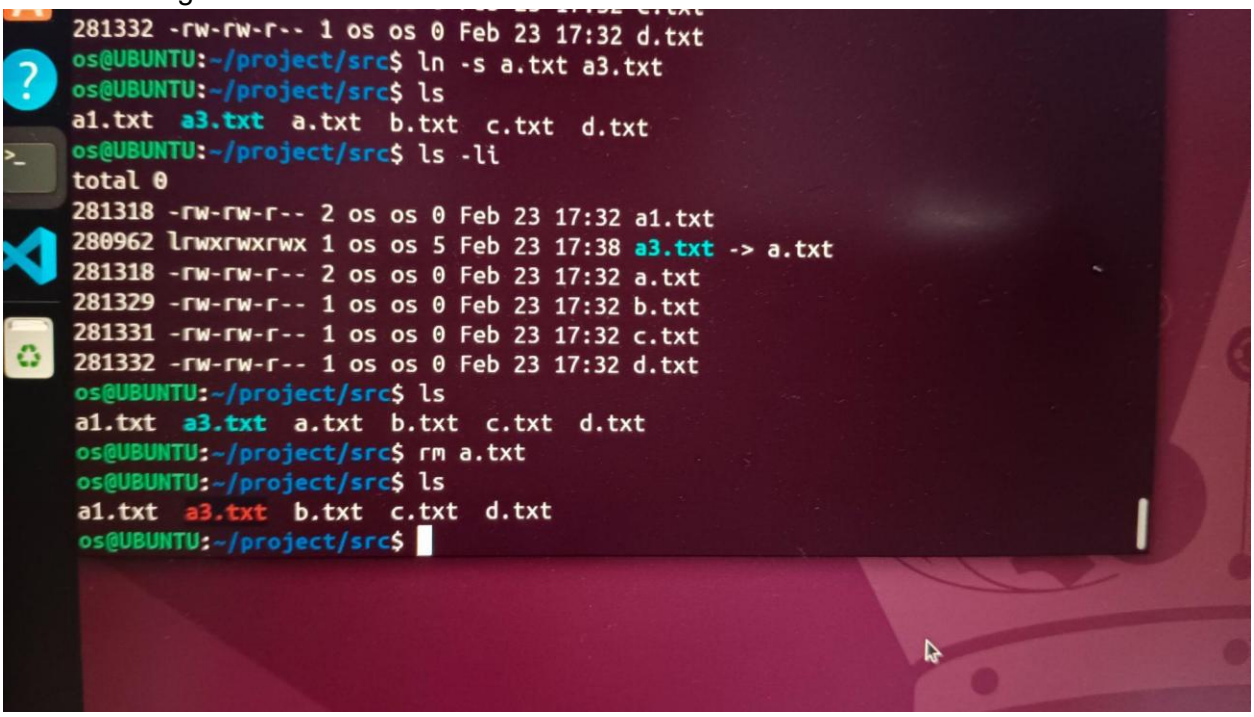
   a. One hard link



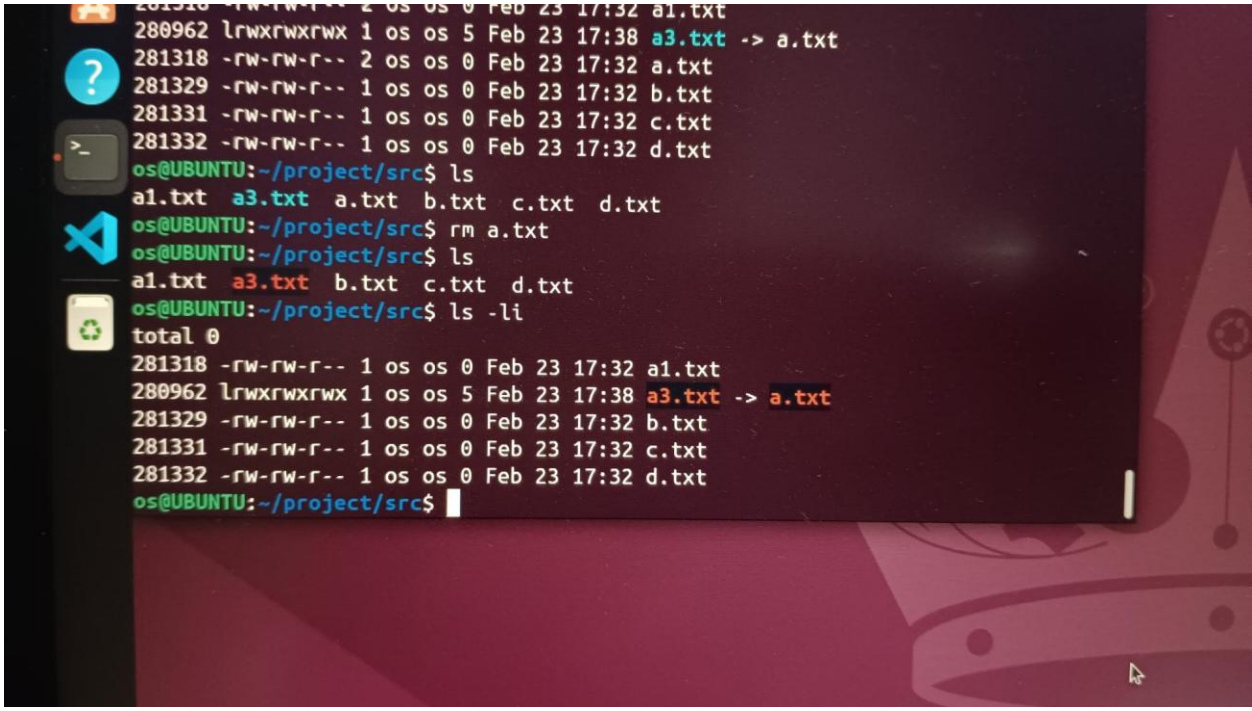
b. One symbolic link



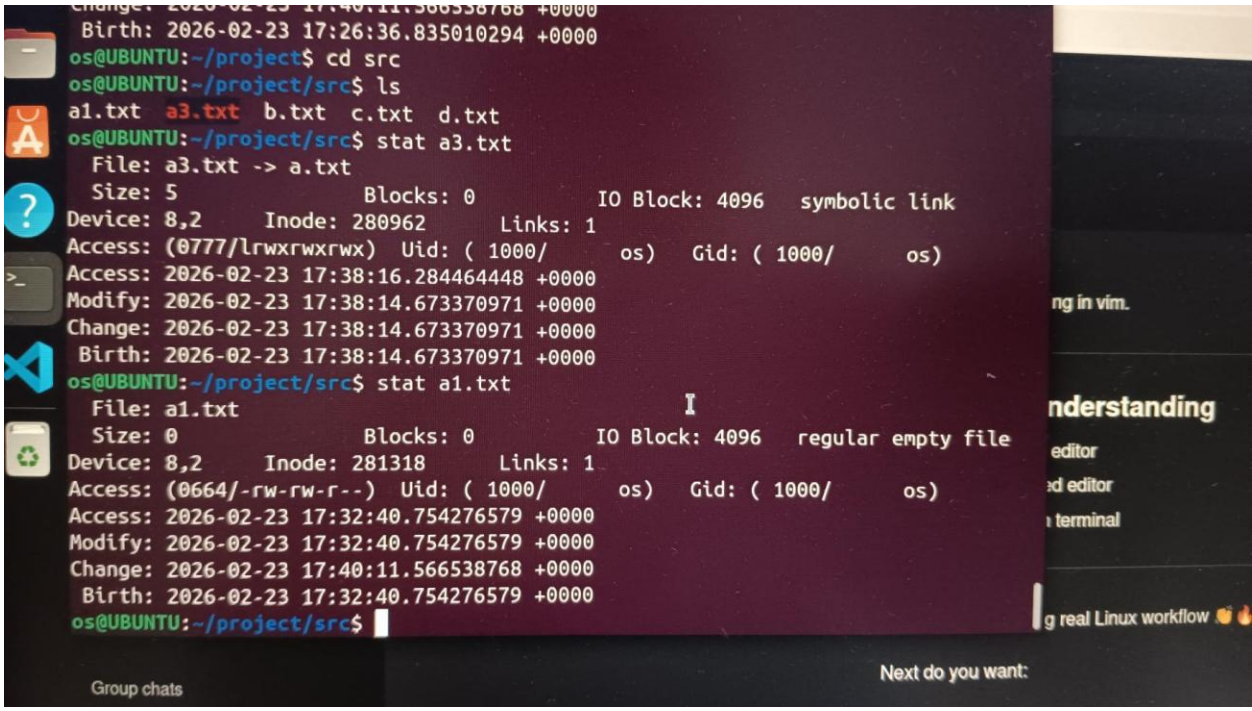
4. Delete the original file.



1. Observe behavior using:
    - `ls -li`


281318 -rw-rw-r-- 2 os os 0 Feb 23 17:32 a1.txt
280962 lrwxrwxrwx 1 os os 5 Feb 23 17:38 a3.txt -> a.txt
281318 -rw-rw-r-- 2 os os 0 Feb 23 17:32 a.txt
281329 -rw-rw-r-- 1 os os 0 Feb 23 17:32 b.txt
281331 -rw-rw-r-- 1 os os 0 Feb 23 17:32 c.txt
281332 -rw-rw-r-- 1 os os 0 Feb 23 17:32 d.txt
os@UBUNTU:~/project/src$ ls
a1.txt a3.txt a.txt b.txt c.txt d.txt
os@UBUNTU:~/project/src$ rm a.txt
os@UBUNTU:~/project/src$ ls
a1.txt a3.txt b.txt c.txt d.txt
os@UBUNTU:~/project/src$ ls -li
total 0
281318 -rw-rw-r-- 1 os os 0 Feb 23 17:32 a1.txt
280962 lrwxrwxrwx 1 os os 5 Feb 23 17:38 a3.txt -> a.txt
281329 -rw-rw-r-- 1 os os 0 Feb 23 17:32 b.txt
281331 -rw-rw-r-- 1 os os 0 Feb 23 17:32 c.txt
281332 -rw-rw-r-- 1 os os 0 Feb 23 17:32 d.txt
os@UBUNTU:~/project/src$

- stat


Birth: 2026-02-23 17:26:36.835010294 +0000
os@UBUNTU:~/project$ cd src
os@UBUNTU:~/project/src$ ls
a1.txt a3.txt b.txt c.txt d.txt
os@UBUNTU:~/project/src$ stat a3.txt
File: a3.txt -> a.txt
Size: 5 Blocks: 0 IO Block: 4096 symbolic link
Device: 8,2 Inode: 280962 Links: 1
Access: (0777/lrwxrwxrwx) Uid: (1000/ os) Gid: (1000/ os)
Access: 2026-02-23 17:38:16.284464448 +0000
Modify: 2026-02-23 17:38:14.673370971 +0000
Change: 2026-02-23 17:38:14.673370971 +0000
Birth: 2026-02-23 17:38:14.673370971 +0000
os@UBUNTU:~/project/src$ stat a1.txt
File: a1.txt
Size: 0 Blocks: 0 IO Block: 4096 regular empty file
Device: 8,2 Inode: 281318 Links: 1
Access: (0664/-rw-rw-r--) Uid: (1000/ os) Gid: (1000/ os)
Access: 2026-02-23 17:32:40.754276579 +0000
Modify: 2026-02-23 17:32:40.754276579 +0000
Change: 2026-02-23 17:40:11.566538768 +0000
Birth: 2026-02-23 17:32:40.754276579 +0000
os@UBUNTU:~/project/src$
Group chats
Next do you want:

Q. Explain inode behavior

Ans. inode is assigned to each file and it is unique for each file.
But in the hard link we can see that 2 files have the same inode number. This was because the inode number is always one but it's name can be more that one or two and that name is pointed to the same inode.

Q. Explain why one link survives and one breaks

Ans. One link Survives and one breaks because:-

1. Hard link survive because when we create a copy from the parent then copy was not pointed to parent it was pointed to inode that's why when parent was deleted it was not broken
2. In Symbolic link the new file that was broke when it's parent was deleted as in symbolic link the copy file is not pointed to inode it was pointed to it's parent that's why it broke.

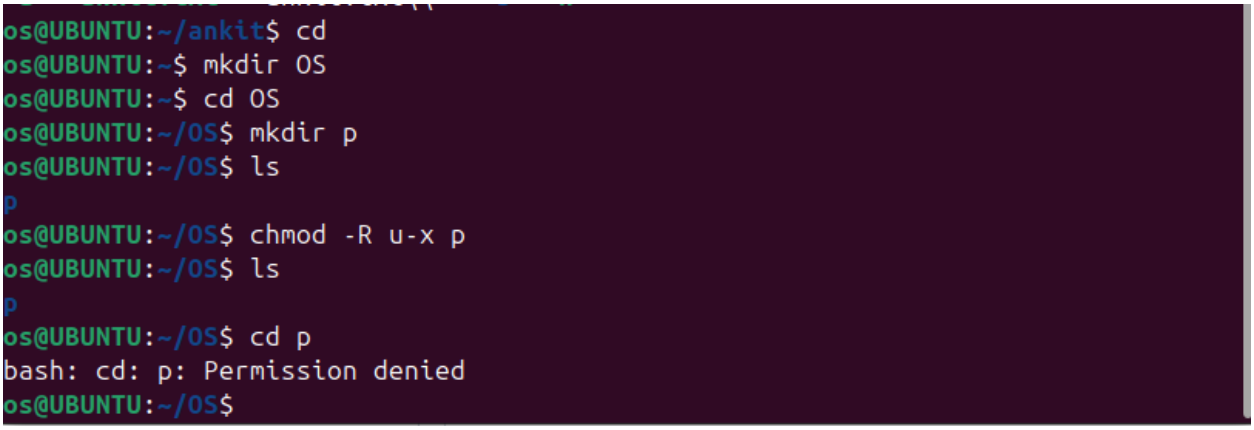
Assignment 2

You are given:

```
Permission denied
```

Simulate 3 different causes:

- Missing execute on directory



In this I remove the permission for execute on P folder

```
p
os@UBUNTU:~/OS$ chmod -R u-x p
os@UBUNTU:~/OS$ ls
p
os@UBUNTU:~/OS$ cd p
bash: cd: p: Permission denied
os@UBUNTU:~/OS$ chmod -R u+x p
os@UBUNTU:~/OS$ cd p
os@UBUNTU:~/OS/p$ ls p
ls: cannot access 'p': No such file or directory
os@UBUNTU:~/OS/p$ ls
os@UBUNTU:~/OS/p$
```

Regive the permission to execute p

- Wrong ownership

```
os@UBUNTU:~/OS/p$ ls
owner
os@UBUNTU:~/OS/p$ nano owner
os@UBUNTU:~/OS/p$ ls
owner
os@UBUNTU:~/OS/p$ chmod 000 owner
chmod: changing permissions of 'owner': Operation not permitted
os@UBUNTU:~/OS/p$ sudo chown os owner
os@UBUNTU:~/OS/p$ ls
owner
os@UBUNTU:~/OS/p$ chmod 000 owner
os@UBUNTU:~/OS/p$
```

In this I change the owner to root and then re back ownership to os user

- Incorrect file mode

```
os@UBUNTU:~/OS/p$ sudo chown os owner
os@UBUNTU:~/OS/p$ ls
owner
os@UBUNTU:~/OS/p$ chmod 000 owner
os@UBUNTU:~/OS/p$ cat owner
cat: owner: Permission denied
os@UBUNTU:~/OS/p$ nano owner
os@UBUNTU:~/OS/p$ chmod 777 owner
os@UBUNTU:~/OS/p$ cat owner
XHello everyone

os@UBUNTU:~/OS/p$
```

Change the permission of owner and then reback it

Then fix them without using `sudo` unless necessary.

Sudo was only used in changing the ownership.

🎯 Assignment 3: Safe Backup Workflow

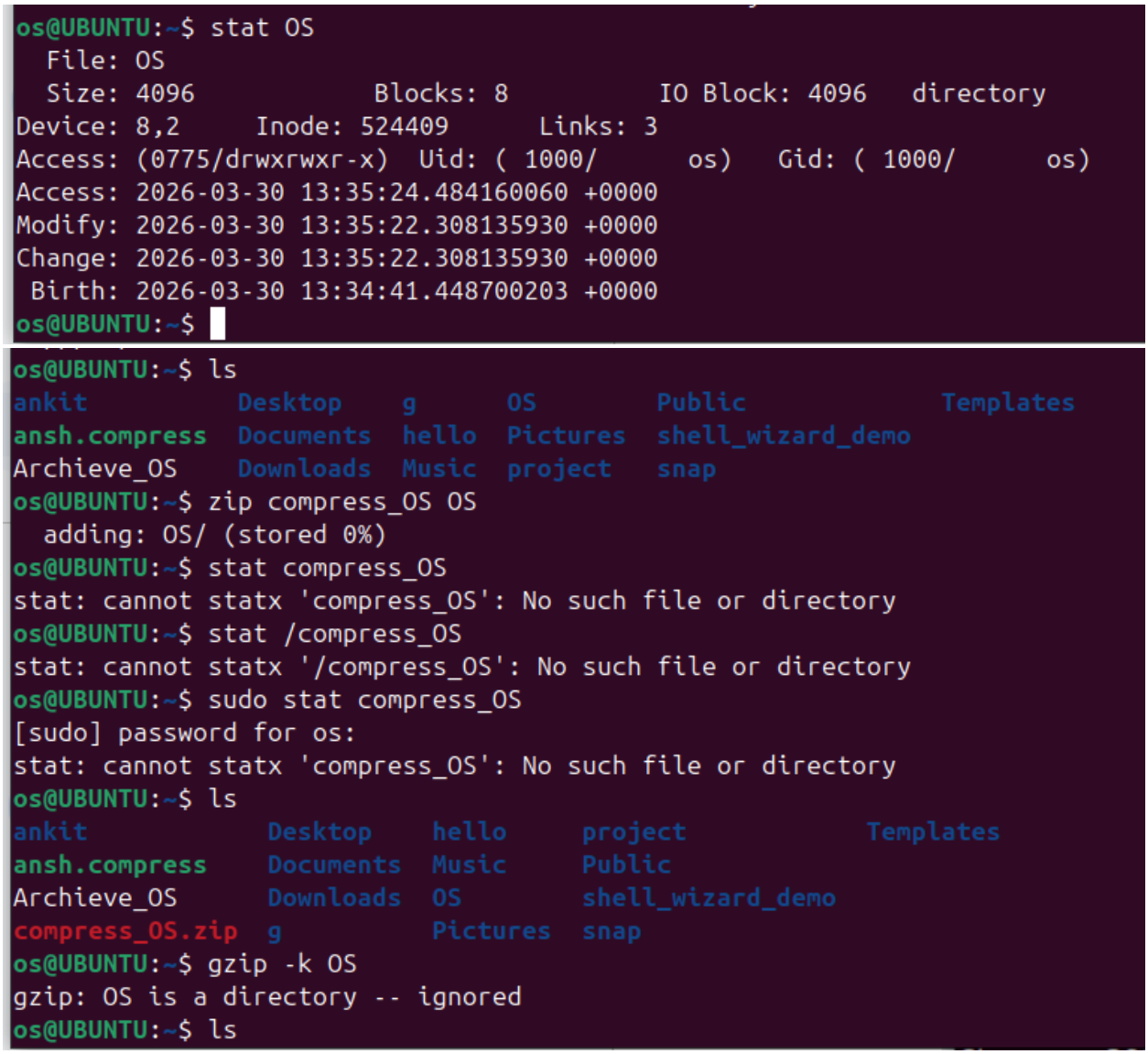
Create a script-less manual workflow that:

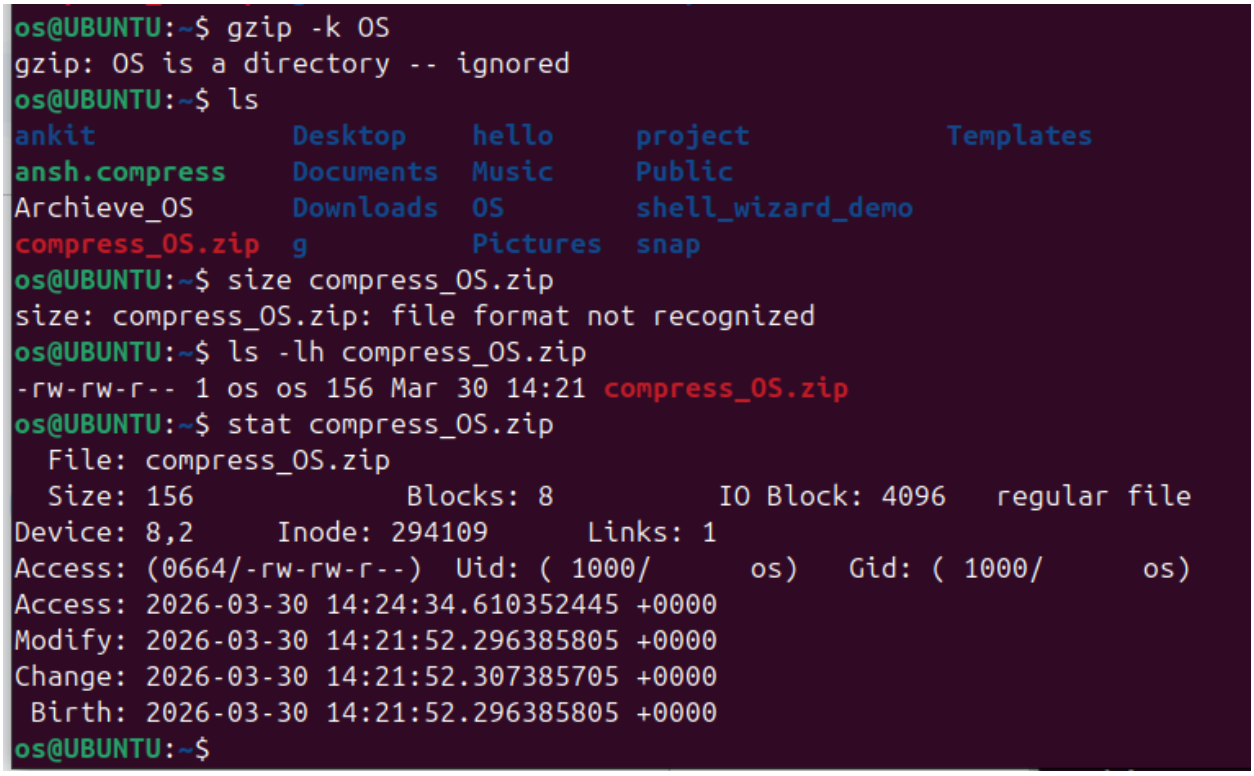
- Archives a directory

```
os@UBUNTU:~$ rm Compress_OS
os@UBUNTU:~$ l
ankit/          Documents/  hello/  Pictures/  shell_wizard_demo/
ansh.compress*  Downloads/  Music/  project/   snap/
Desktop/        g/          OS/     Public/    Templates/
os@UBUNTU:~$ ls
ankit          Documents  hello  Pictures  shell_wizard_demo
ansh.compress  Downloads  Music  project   snap
Desktop        g          OS     Public    Templates
os@UBUNTU:~$ tar cvf Archieve_OS OS
OS/
OS/p/
OS/p/owner
OS/p/gh/
OS/p/hf/
os@UBUNTU:~$ ls
ankit          Desktop    g      OS        Public             Templates
ansh.compress  Documents  hello  Pictures  shell_wizard_demo
Archieve_OS    Downloads  Music  project   snap
os@UBUNTU:~$
```

IN this I archive the OS folder by the name Archive_OS

- Compresses it





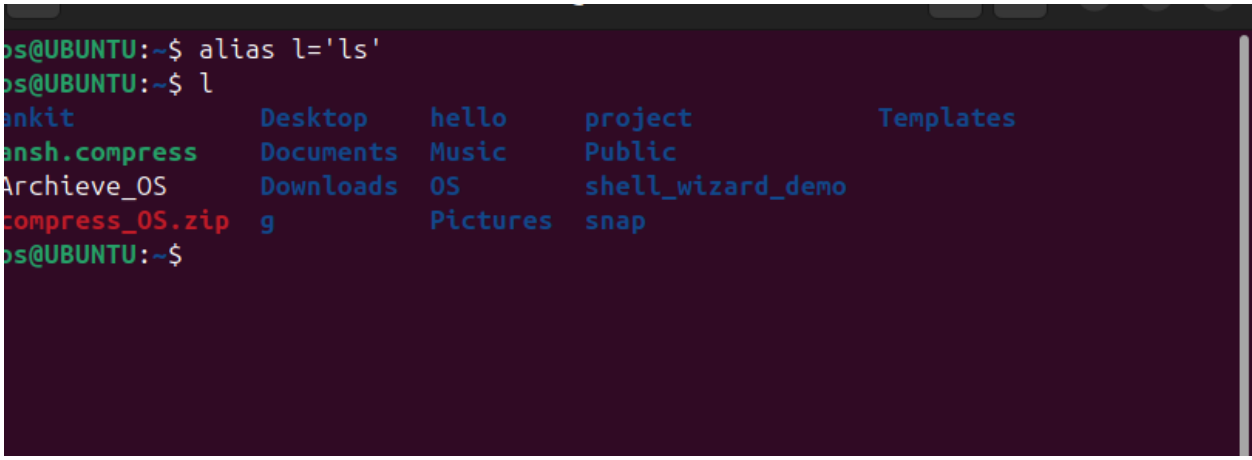
In this I make compress the OS and compressed file name compress_OS.zip

- Stores it with timestamp
- Verifies size before and after

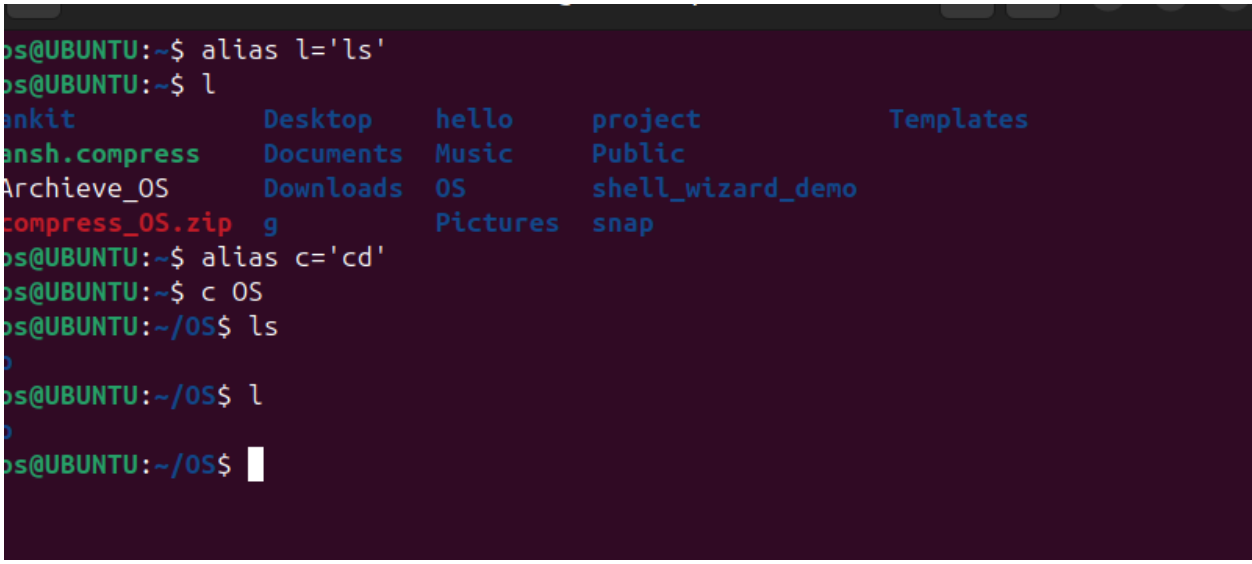
🎯 Assignment 4: Shell Personalization & Alias Behavior

Task:

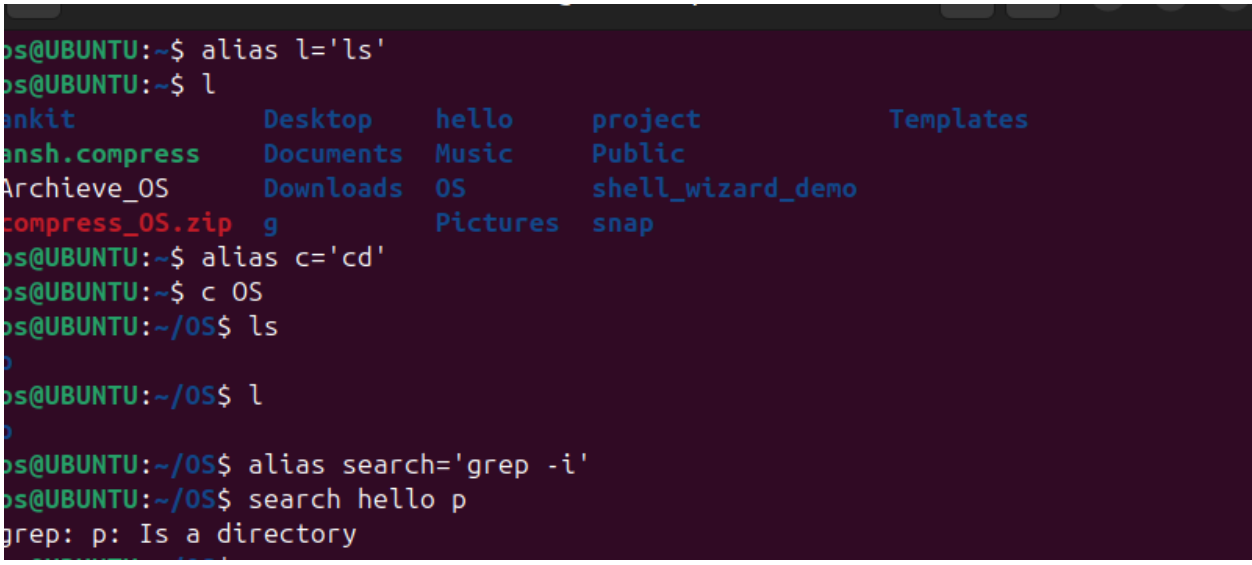
1. Create 5 useful aliases:
    - One for `ls`

- 

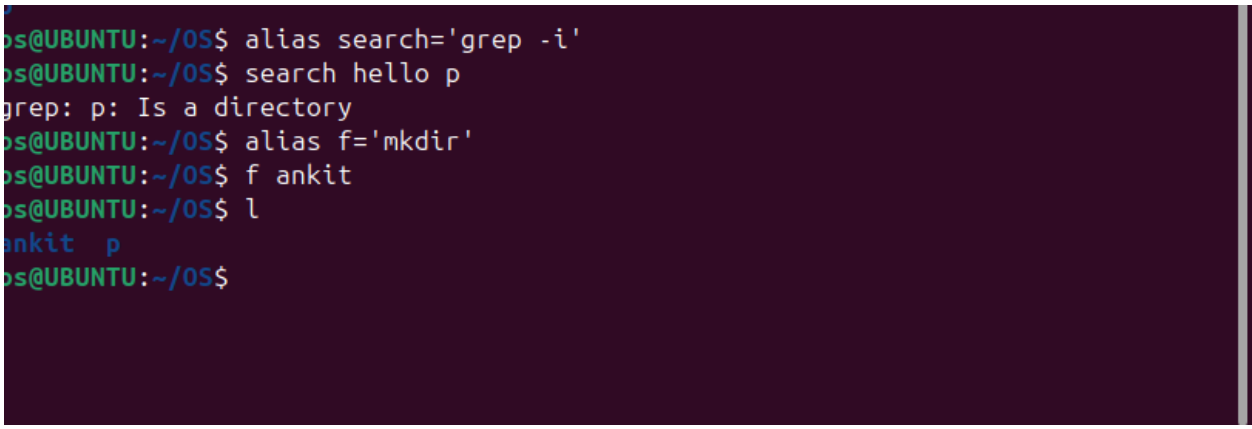
- One for navigation



- One for search



- One for git (if installed)
- One custom productivity shortcut



2. Make them:
   - Work in current session
   - Persist across terminal restarts

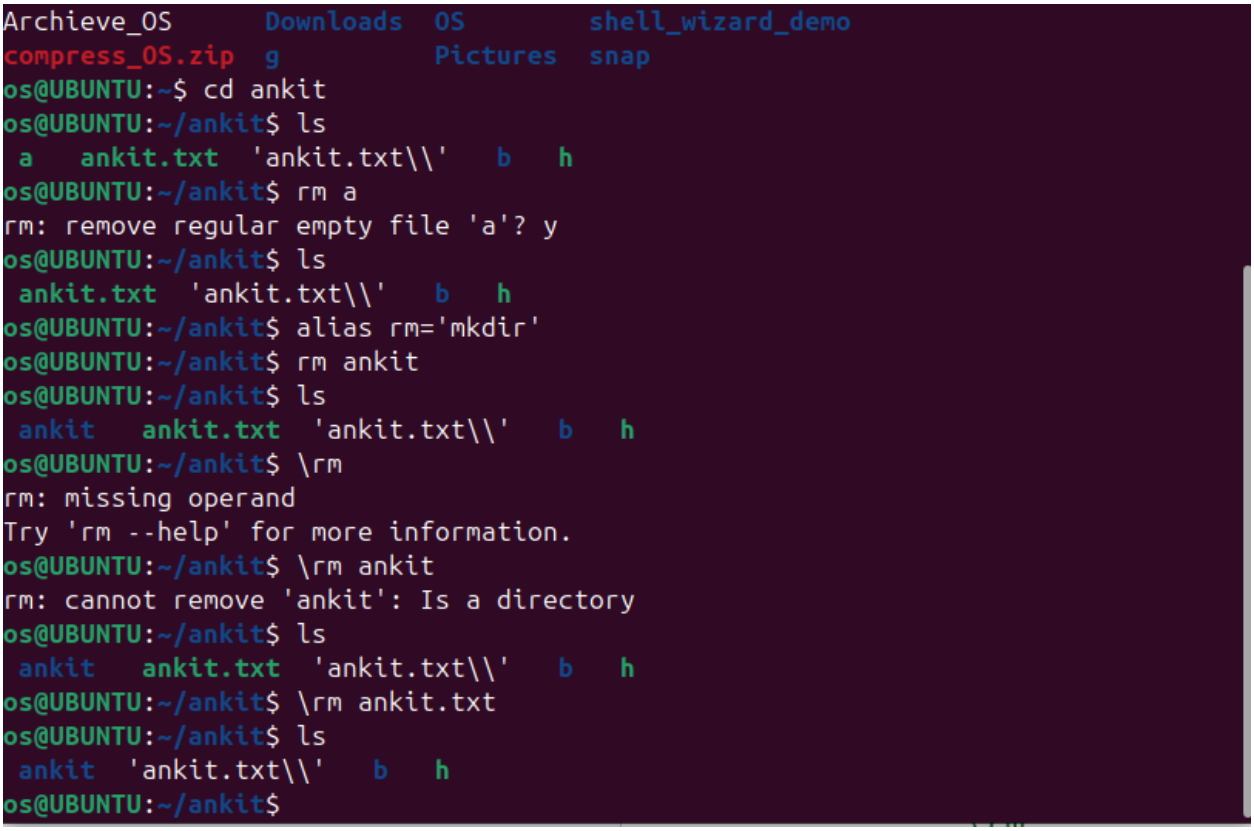
3. Verify:
    - `type <alias_name>`
    - `alias`
    - Restart terminal and test again

- Create an alias that breaks something intentionally.
  Example:

  ```
  alias rm='rm -i'
  ```

- Then override it temporarily using:

  ```
  \rm
  ```

- 

Deliverable:

- Explain why `\command` bypasses alias

  I think when we use '\' this then system go to root process.

- Explain why aliases don't work inside non-interactive scripts
- Show difference between alias and function